SEMINAR PRESENTATION

Gestational Diabetes Mellitus

Diagnosis, complications and the Ayurvedic management of madhumeha in pregnancy

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Venue
Department of Prasuti Tantra & Stree Roga seminar series

Year
2023

> *The diagnostic threshold for gestational diabetes changed in 2025. This seminar covers the screening, the surveillance and the classical alternatives.*

Abstract

Gestational diabetes mellitus is now the commonest medical complication of pregnancy and carries substantial maternal and fetal consequences, including macrosomia, hypertensive disorders and neonatal hypoglycaemia. This seminar covers the screening and diagnostic criteria, the glucose targets and monitoring required, the nutritional and lifestyle management of the condition, the pharmacological options, the intrapartum and postpartum management, and the long-term risk of progression to type 2 diabetes. Alongside the modern pathway, the presentation discusses the Ayurvedic understanding of madhumeha in pregnancy — the doshic basis, the classical ahara-vihara prescription and the role of herbo-mineral formulations — and evaluates where classical and contemporary management can be integrated responsibly.

Gestational diabetes mellitus · GDM · Madhumeha · Prasuti Tantra · Ayurveda · Metabolic health · Antenatal care · Macrosomia

Cite this work

APA 7TH

Dr. Nidhi Sanjay Patel (2023). Gestational Diabetes Mellitus [Seminar presentation]. Department of Prasuti Tantra & Stree Roga seminar series.

VANCOUVER

1. Dr. Nidhi Sanjay Patel. Gestational Diabetes Mellitus [seminar presentation]; 2023; Department of Prasuti Tantra & Stree Roga seminar series.

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Topic of Presentation

DIABETES MELLITUS IN PREGNANCY (GDM)

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INTRODUCTION:

Diabetes mellitus is a chronic metabolic disorder due to either insulin deficiency (relative or absolute) or due to peripheral tissue resistance (decrease sensitivity) to the action of insulin.

The pathophysiology involved are:

Decreased sensitivity of skeletal muscles and liver to insulin (insulin resistance) and

Inadequate secretion of insulin (β cell dysfunction).

Pregnancy is a state of chronic low grade inflammation.

This is associated with increased circulating levels of C-reactive protein (CRP) and interleukin-6 (IL-6). Both these factors enhance insulin resistance. The defect lies both in insulin secretion and action. The ultimate effect is the hyperglycemia.

Globally, one in 10 pregnancies is associated with diabetes, 90% of which Gestational Diabetes mellitus. Undiagnosed or inadequately treated GDM can lead to significant maternal and foetal complications.

Maternal carbohydrate metabolism is generally altered in pregnancy by the influence of pregnancy hormones to transfer of glucose from the mother to the foetus which is needed throughout pregnancy.

Hormones like Human Placental lactogen, oestrogen, Progesterone and cortisol, prolactin, free fatty acids, leptin are responsible for increased tissue resistance to insulin due to continuous supply of glucose to foetus.

CLASSIFICATION OF DIABETES IN PREGNANCY:

In Pregnancy, the diabetes can be grouped into two categories:

Overt or pregestational diabetes:

This is seen in women known to be diabetic before the onset of pregnancy. They may have absolute insulin deficiency (Type- I) or defective insulin secretion or insulin resistance (Type-II).

Gestational diabetes:

GDM is defined as carbohydrate intolerance of variable severity with onset or first recognition during the present pregnancy.

GESTATIONAL DIABETES MELLITUS (GDM)

GDM is defined as carbohydrate intolerance of variable severity with onset or first recognition during the present pregnancy. The entity usually presents late in the second or during the third trimester.

Risk Factors of Diabetes in pregnancy:

Positive family history of diabetes.

Having a Previous birth of an overweight baby of > 4 kgs

Previous stillbirth or unexplained foetal loss.

Presence of polyhydramnios or recurrent vaginal candidiasis in present pregnancy.

Persistent glycosuria.

Age over 30 years.

Obesity.

SCREENING:

Screening strategy for detection of GDM are:

(a) Low risk—Absence of any risk → blood glucose testing is not routinely required

(b) Average risk—Some risk factors → perform screening test

(c) High risk—Blood glucose test as soon as feasible.

The method employed is by using 50 gm oral glucose challenge test without regard to time of day or last meal, between 24 weeks and 28 weeks of pregnancy.

A plasma glucose value of 140 mg% or that of whole blood of 130 mg% at 1 hour is considered as cut off point for consideration of a 100 gm (WHO– 75 gm) glucose tolerance test.

DIPSI (Diabetes In Pregnancy Societies of India): recommends 1-step procedure with 75 gm oral glucose without regard to the time of the last meal. A venous plasma glucose value at 2-hour more than 140 mg/dL is diagnosed GDM.

Hazards of GDM:

Increased perinatal loss is associated with fasting hyperglycaemia.

Increased incidence of macrosomia.

Polyhydramnios.

Birth trauma.

Recurrence of GDM in subsequent pregnancies is about 50 %.

EFFECTS OF PREGNANCY ON DIABETES

It is difficult to stabilize the blood glucose during pregnancy due to altered carbohydrate metabolism and an impaired insulin action.

The insulin antagonism is due to the combined effect of human placental lactogen, estrogen, progesterone, free cortisol and degradation of the insulin by the placenta.

The insulin requirement during pregnancy increases as pregnancy advances.

As more glucose leaks out in the urine due to renal glycosuria, control of insulin dose cannot be made by urine test and repeated blood glucose estimation becomes mandatory.

With the “accelerated starvation” concept, there is rapid activation of lipolysis with short period of fasting.

Ketoacidosis can be precipitated during hyperemesis in early pregnancy, infection and fasting of labor.

It can be iatrogenically induced by β sympathomimetics and corticosteroids used in the management of preterm labor.

Insulin requirement falls significantly in puerperium.

Vascular changes, especially retinopathy, nephropathy, coronary artery disease and neuropathy may be worsened during pregnancy

Effects of Diabetes on pregnancy:

MATERNAL

FETAL

During Pregnancy

During Labour

During Puerperium

Miscarriage.

Preterm labour.

Infection

Increased incidence of pre-eclampsia.

Polyhydramnios.

Maternal distress

Diabetic retinopathy

Prolongation of labour due to big baby,

Shoulder dystocia.

Perineal injuries.

Postpartum haemorrhage

Operative interference.

Puerperal sepsis.

Lactation failure.

Foetal macrosomia.

Congenital malformation

Growth restriction Foetal death

Major Birth Defects in Infants of Diabetic Mothers (6–10%)

CNS And Skeletal

Cardiovascular

Renal

Gastrointestinal

Others.

Neural Tube defects.

Anencephaly.

Microcephaly.

Caudal regression syndrome.

Sacral agenesis

Ventricular septal defect, Arterial septal defect.

Coarctation of aorta

Transposition of great vessels.

Renal agenesis

Hydronephrosis

Polycystic kidneys.

Double ureter.

Duodenal atresia

Anorectal atresia

Omphalocele

Tracheo esophageal fistula

Single umbilical artery.

Neonatal complications: Hypoglycaemia, Respiratory distress syndrome, Hyperbilirubinemia, Cardiomyopathy.

Long term effects:

Childhood obesity, metabolic syndrome, neuropsychological effects, and juvenile diabetes.

Investigations:

Repeat and random Urine samples taken on one or more occasions throughout pregnancy reveal glycosuria in about 5-50% cases.

Persistent glycosuria is an indication of OGTT.

Glycosylated Haemoglobin should be determined at the end of 1st trimester and trimonthly thereafter. HbA1c level of <6 is desirable.

Sonographic evaluation in pregnancy at 3-4weeks interval is extremely helpful, not only to diagnose varieties of congenital malformation of foetus but also to detect foetal macrosomia.

OVERT DIABETES

A patient with symptoms of diabetes mellitus (polyuria, polydipsia, weight loss) and random plasma glucose concentration of 200 mg/dL or more is considered overt diabetic.

The condition may be pre-existing or detected for the first time during present pregnancy.

According to American Diabetic Association diagnosis is positive if:

(a) the fasting plasma glucose exceeds 126 mg/dL,

(b)the 2 hours post glucose (75 gm) value exceeds 200 mg/dL and

(c) HbA1C ≥ 6.5%

MANAGEMENT

Preconceptional counseling

Principles in the management:

Careful antenatal supervision and glycaemic control.

To find out the optimum time and method of delivery.

Arrangement for care of the new-born.

ANTENATAL CARE:

Antenatal supervision should be at monthly intervals up to 20 weeks and thereafter at 2 weeks intervals.

At times patient needs admission for stabilization of blood glucose and for monitoring the fetus.

Diet—30 Kcal/kg for normal weight women, 24 Kcal/kg for overweight women and 12 Kcal/kg for morbidly obese women.

Ideal plasma glucose (mg/dL) levels in women with pregestational diabetes should be: fasting < 95, premeal < 100; 2 hour P.P. values < 120 and morning (2 am–6 am) value > 60.

If values are exceeded even on diet, insulin therapy is suggested.

Diet should contain carbohydrate 40%–50%, protein 20%, fat 30–40% and saturated fat <10%.

Fat may be curtailed, if the patient is obese.

Fiber containing diet (complex carbohydrates) is increased.

Usually three meal regimen, with breakfast 25% of the total calorie intake, lunch 30%, dinner 30% and several snacks are quite suitable for most of the patients.

Frequent blood sugar estimation is required as the urine examination for sugar is not informative.

Monitoring of blood glucose by glucose meter can give an accurate idea about the control.

In addition, glycosylated hemoglobin should be determined at the end of first trimester and trimonthly thereafter. HbA1c level of ≤6% is desirable.

Sonographic evaluation in pregnancy (at 3-4 weeks interval) is extremely helpful, not only to diagnose varieties of congenital malformation of fetus but also to detect fetal macrosomia or growth restriction (rare).

Assessment of fetal well-being is to be made from 28 weeks onwards.

Biophysical profile and NST should be performed weekly. Doppler umbilical artery velocimetry is useful in cases with vasculopathy.

Insulin therapy:

When diabetes is first detected during pregnancy and cannot be controlled by diet alone, it should be treated with insulin.

A postprandial (2 hours) plasma glucose level of more than 140 mg% even on diet control is an indication of insulin therapy.

There is frequent change in insulin need during pregnancy and changes in the dosage are made in small increments at a time.

Glycemic goals should be around 90 mg/dL before meals and not to exceed 120 mg/dL, 2 hours after meals.

During the stabilization process of the insulin dose, frequent blood sugar estimation especially at night (2 am–6 am) may be necessary using glucose meter (capillary whole blood, utilized for self-monitoring, is equivalent to venous plasma).

However, as pregnancy advances, “a double mixed regime” may be employed.

The patient should receive three to four daily injections of a regular (human act rapid) and an intermediate acting insulin (isophane), the latter is to be given before dinner.

The aim is to maintain the blood sugar level as near to normal as possible without causing troublesome hypoglycemia.

Use of subcutaneous insulin infusion by insulin pump is preferred as it is more physiological.

Women are instructed on diet composition, insulin dose, recognition and treatment of hypoglycemia, hyperglycemia and ketosis, adjusting insulin dose in relation to exercise, food and sick days.

Oral hypoglycemic drugs have been found to be effective and safe. Commonly used drugs are glibenclamide and metformin (biguanide).

Both the drugs cross the placenta. However, no teratogenic effect has been observed as yet.

Glibenclamide appears to be superior when compared to metformin.

ADMISSION:

In uncomplicated cases, the patient is admitted at 34–36 weeks.

Early hospitalization facilities:

(1) Stabilization of diabetes

(2) Minimizes the incidence of preeclampsia, polyhydramnios and preterm labor

(3) To select out the appropriate time and method of delivery.

Induction of labor:

The indications are—

Diabetic women controlled on insulin (GDM or class B diabetes) are considered for induction of labor after 38 completed weeks

Women with vascular complications (preeclampsia, IUGR) often require induction after 37 weeks.

Methods:

Prior to the day of induction of labor, the usual bed time dose of insulin is administered.

No breakfast and no morning dose of insulin are given on the day of induction.

Capillary blood glucose level is checked with a bed side glucose meter.

Normal saline infusion is begun.

Induction is done by low rupture of the membranes. Simultaneous oxytocin drip is started, if not contraindicated.

An intravenous drip of one liter of 5% dextrose is set up with 10 units of soluble insulin.

An infusion rate of 100–125 mL/ hr (1–1.25 units/hr), will maintain a good glucose control to approximately 100 mg/dL (ACOG-2005).

Insulin may also be infused from a syringe pump (0.25–2 units/hr).

Blood glucose levels are estimated hourly with a glucose meter and the soluble insulin dose is adjusted accordingly.

Epidural analgesia is ideal for pain relief. If the labor fails to start within 6–8 hours or if the labor progresses unsatisfactorily, cesarean section should be performed.

Cesarean section:

The indications are—

(1) Fetal macrosomia (>4 kg)

(2) Diabetes with complications or difficult to control

(3) Fetal compromise as observed in antepartum fetal monitoring

(4) Elderly primigravidae

(5) Multigravidae with a bad obstetric history

(6) Obstetric complications like preeclampsia, polyhydramnios, malpresentation.

As such 50% of diabetic mothers are delivered by cesarean section.

PUERPERIUM:

Antibiotics should be given prophylactically to minimize infection.

Insulin requirement falls dramatically following delivery.

She is to revert to the insulin regime as was prior to pregnancy.

A fresh blood glucose level after 24 hours will help to adjust the dose of insulin.

Breastfeeding is encouraged. Women who breast feed should have additional 500 Kcal daily in diet.

In lactating women insulin dose is lower.

CARE OF THE BABY:

A neonatologist should be present at the time of delivery. The baby should preferably be kept in an intensive neonatal care unit and to remain vigilant for at least 48 hours, to detect and to treat effectively any complication likely to arise.

Asphyxia is anticipated and be treated effectively.

To look for any congenital malformation.

All babies should have blood glucose to be checked within 2 hours of birth to avoid problems of hypoglycemia (blood glucose < 35 mg/dL).

All babies should receive 1 mg vitamin K intramuscularly.

Early breastfeeding within half to 1 hour is advocated and to be repeated at three to four hourly intervals thereafter to minimize hypoglycemia and hyperbilirubinemia.

Improvement in the care of diabetes in pregnancy has reduced perinatal mortality significantly (< 5%).

According to Ayurveda,

GARBHINI PRAMEHA

Direct references of Garbhini Prameha (Gestational Diabetes Mellitus) is not described in Bruhat Trayee and Laghu Trayee.

निरुक्ति (Nirukti)

मधुरं यच्च सर्वेषु प्रायो मध्विव मेहति। सर्वेऽपि मधूमेहाख्या माधुर्याच्च तनोरतः।।

(अ.ह.नि. 10/21)

In madhumeha, urine of the patient becomes sweet like that of honey, and whole body becomes sweet, hence called so.

क्षारेण किंवा कटुना कषायेण वर्जितैर्गर्भवतीजनस्य।

आजेस्करैर्वा मधुरैः सदैवान्नपान जातैरतिमात्र भक्षितैः ।। (Bhaishajya ratnavali Cha 90/3)

In Bhaisajya Ratnawali, it is mentioned that excessive use of atimadhura, atiguru, paryushita ahara & atibhojna and completely neglecting the kshara, katu as well as kashaya rasa dravyas during pregnancy will lead to ojomeha in garbhini.

Acharyas have advised brumhana for garbhini and indicated hrudya, drava, madhura, snigdha aahara for her.

All these increases kapha dosha which is also the main responsible dosha for madhumeha.

Dushit kapha leads to dushti of dhatus and updhatus resulting in maternal and fetal complications.

So, if garbhini takes kapha vardhak aahara vihaar other than indicated or prakriti virudh aahaar she may be diseased by madhumeha.

Nidana

आहारज – kaphakara ahara- guru, snigdha, sandra, picchila, sheet, madhura ahara atisevana.

Navannapana, sheetal & dushit jala, ikshu, guda, dadhi atisevana,

Viruddha bhojana, paryushitahara.

विहारज- Divaswapna, Atihasyam, Alashya, Atisnehana.

मानसिक- Chinta, Shoka, Bhaya.

अन्यज - Bija bhaga ayvaya dushti , Abhighataja

Samprapti:-

633730278701500

Lakshanas

Effect of varjya ahara & vihara sewana on garbhini in Garbhavastha:

Garbha ativriddhi, Garbhodaka ativriddhi, Mutra ati pravratti, Yoni kandu, Yoni srava.

Chikitsa :-

Pre-conceptional care mentioned by Acharyas :-

Acharya have mentioned snehana, swedana, vamana, virechana & basti karma for purification of body.

By proper purification and samskaras, unvitiated yoni, garbhasaya, manas, and beeja are ensured leading to healthy pregnancy.

Garbhini paricharya:-

Acharya kasyapa says that the ahara sevan by the pregnant woman become satmya to the fetus.

Thus, the aahara should be taken considering desha, kala, and digestive capacity.

Chikitsa sidhdhant for Garbhini Prameha :-

Nidaana parivarjana

Ama pachana chikitsa

Deepana pachana yogas are:

Pippali ksheera paka, Jeeraka kashaya, Yavagu, peya processed with deepaniya drugs.

आभ्यन्तर प्रयोग (Internally)

Churṇa of jambu phala majja (4 part), nimba patra (4 part), amalaki (4 part), bilwa (2 part), karavelaka (8 part): 10-12 g taken thrice daily.

Asanadi, nyagrodhadi varga, khadiradi kwatha etc. are effective in madhumeha.

Haridra and amalaki kwatha is beneficial.

CONCLUSION

Gestational diabetes is a common problem in India. Risk stratification and screening is essential in all pregnant women. Strict glycemic targets are required for optimal maternal and fetal outcome. Patient education is essential to meet these targets. Long term follow up of the mother and baby is essential. According to Ayurveda efforts of having healthy baby commences with pre-conception care and management. But in India most of the females visit clinics after having pregnancy, so role of preconception care and management is very limited. Garbhini parichrya by monthly regimen described by various acharyas is highly effective for this purpose. If pregnant woman strictly follows Garbhini parichrya and ritumati charya before conception, diseases of garbhini and garbha can be avoided.

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